D2NN vs MPLC

Physics of Multi-Plane Phase Systems
for Atmospheric Turbulence Mitigation

Kim Lab · April 2026

$\lambda = 1.55\ \mu m$ · $C_n^2 = 5\times10^{-14}$ · 100m – 3km · 5-layer D2NN

The Shared Foundation: Unitary Decomposition Theorem

Both D2NN and MPLC implement the same physics: cascaded phase masks with free-space propagation can realize any unitary spatial transformation (Reck et al. 1994, Morizur et al. 2010).

Any unitary $\hat{U}$ can be decomposed as:

$$\hat{U} = \hat{D}_L \cdot \hat{P}(d) \cdot \hat{D}_{L-1} \cdot \hat{P}(d) \cdots \hat{P}(d) \cdot \hat{D}_1$$

$$\hat{D}_\ell = \mathrm{diag}(e^{i\phi_\ell(x,y)})$$

Phase mask (diagonal)

$$\hat{P}(d) = \mathcal{F}^{-1} H(f_x, f_y; d) \mathcal{F}$$

Free-space propagation

$$= \mathrm{e}\hat{}\{\mathrm{ik_z\ d}\},\ = \sqrt{k^2 - k_x^2 - k_y^2}$$

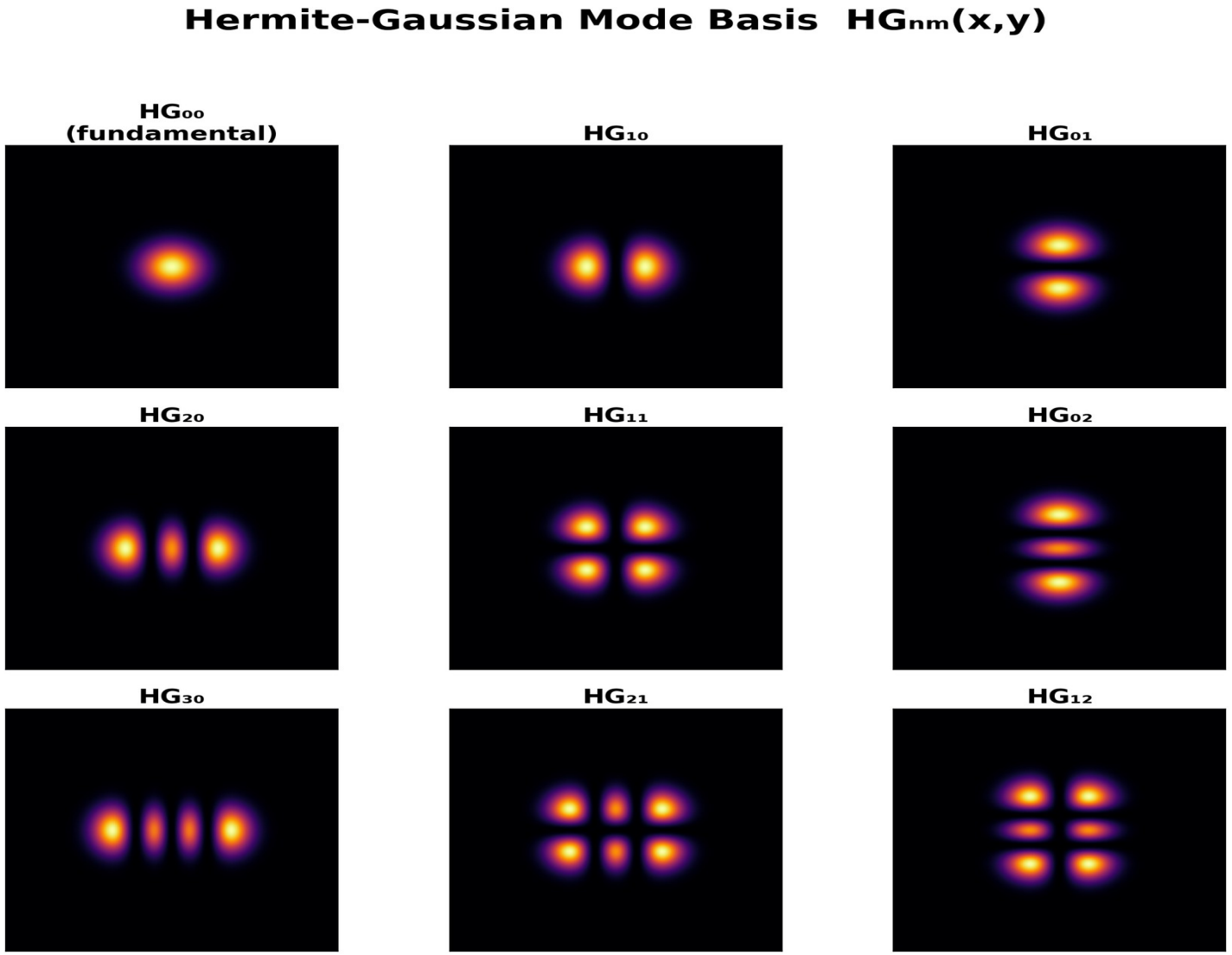
Angular spectrum TF

Key implication: With sufficient layers L, both systems can implement ANY mode transformation — the difference lies in ***how*** the phase masks are determined.

MPLC: masks designed analytically via wavefront matching algorithm

D2NN: masks trained end-to-end via stochastic gradient descent

Hermite-Gaussian Mode Basis



Mode Decomposition

Any beam can be decomposed into orthogonal HG modes:

$$U(x,y) = \Sigma\, c_n\, \psi_n(x,y)$$

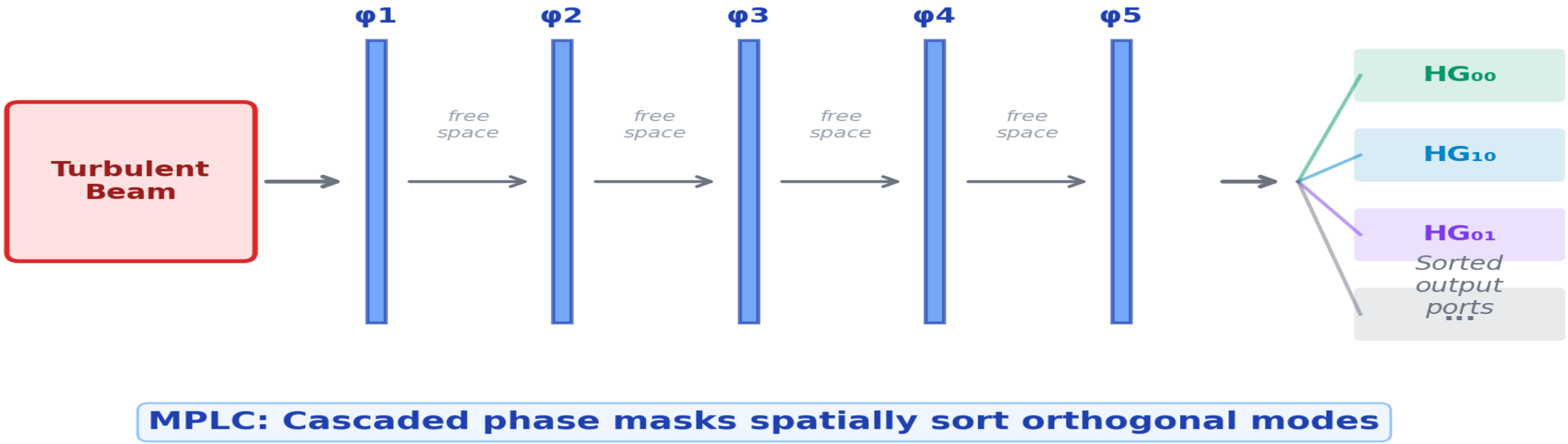
where ψ_n = HG mode function
c_n = complex mode coefficient

Turbulence redistributes energy across modes:

- Weak turbulence: most energy in HG_{00} (fundamental)
- Strong turbulence: energy spreads to many higher-order modes
- Number of significant modes ≈ $(D/r_0)^2$

MPLC sorts these modes spatially.
D2NN ignores mode structure entirely.

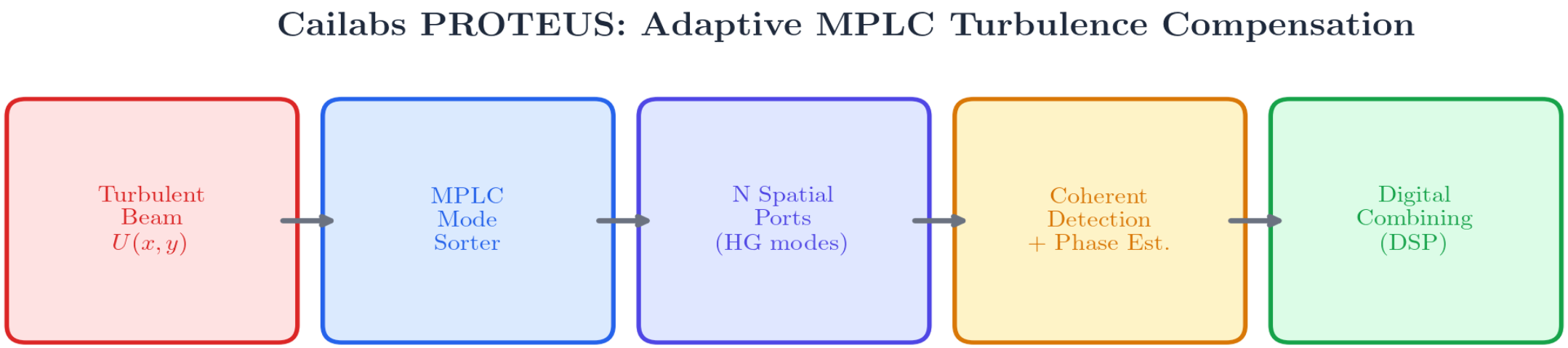
MPLC: Adaptive Mode Sorting Architecture



How it works: Each phase plane φ_l is designed so that after L planes, mode ψ_n emerges at spatial port n. The wavefront matching algorithm (Sakamaki et al.) iteratively optimizes each plane to maximize mode separation.

For turbulence compensation: requires coherent detection at each port + real-time DSP to measure c_n and recombine. This makes it an ***active*** system with significant power consumption and latency requirements.

Cailabs PROTEUS: Commercial MPLC for FSO



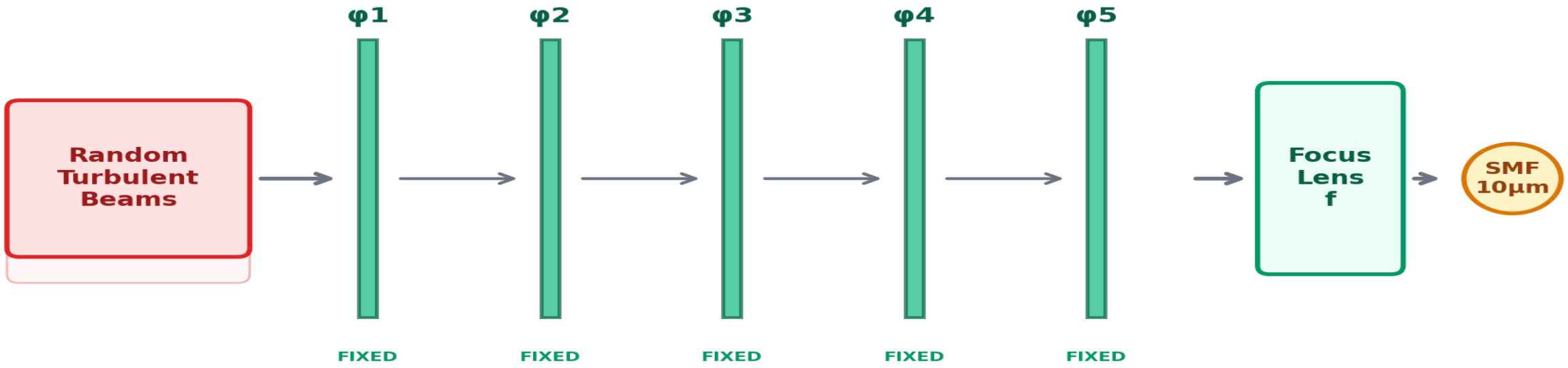
$$U_{\text{turb}}(x, y) = \sum_{n=0}^{N-1} c_n \, \psi_n(x, y)$$

MPLC sorts each ψ_n to port n $\rightarrow$ measure $|c_n|, \angle c_n$ $\rightarrow$ coherently recombine

Requires: wavefront sensor / coherent receiver at each port + real-time DSP

Cailabs approach: MPLC sorts turbulent beam into N HG modes → coherent Rx per port → DSP recombines. Requires SLM or fixed MPLC + N photodetectors + FPGA. Cost: tens of k$.

Our D2NN: Static Ensemble-Optimized Phase Masks



D2NN: Same static masks for ALL realizations → ensemble-optimized bucket power

Loss Function

$$L = -\log(E_{bucket} / E_{input} + \varepsilon)$$

Directly maximizes absolute focal bucket power.
No vacuum reference. Throughput intrinsically preserved.

Key Differences from MPLC

- Same masks for ALL turbulence realizations
- No mode decomposition or sensing
- Zero power consumption (passive)
- **WF RMS unchanged — NOT wavefront correction**
- **Mechanism: energy redistribution + fade floor lifting**

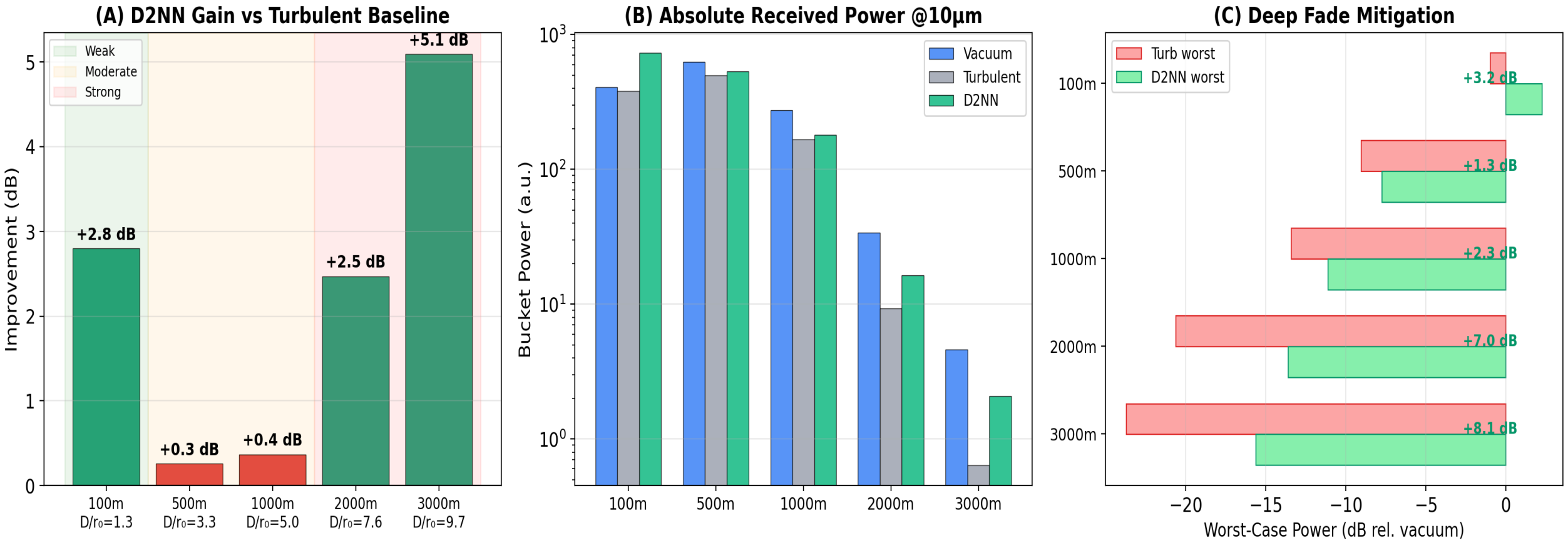
Physics Equations: MPLC vs D2NN

Key Equations: MPLC vs D2NN Physics

Fried parameter:	$r_0 = \left(0.423\, k^2 C_n^2 L\right)^{-3/5}$
Rytov variance:	$\sigma_R^2 = 1.23\, C_n^2\, k^{7/6}\, L^{11/6}$
Mode decomposition:	$U_{\text{turb}} = \sum_{n=0}^{N-1} c_n\, \psi_n^{\text{HG}}(x, y)$
MPLC unitary:	$\hat{U}_{\text{MPLC}} = \prod_{\ell=1}^{L} \hat{D}_\ell \cdot \hat{P}(d)$
D2NN per-layer:	$U_{\ell+1} = \mathcal{P}_d\{U_\ell \cdot e^{i\phi_\ell(x,y)}\}$
Raw RP loss:	$\mathcal{L} = -\log\left(\frac{\sum_{\mathcal{B}} \lvert U_f\rvert^2}{\sum \lvert U_{\text{in}}\rvert^2} + \varepsilon\right)$
PIB (ratio):	$\text{PIB}(r_b) = \frac{\int_0^{r_b} I_f(r)\, r\, dr}{\int_0^{\infty} I_f(r)\, r\, dr}$
Received power:	$\text{RP} = \text{PIB} \times \text{TP} \times E_{\text{in}}$

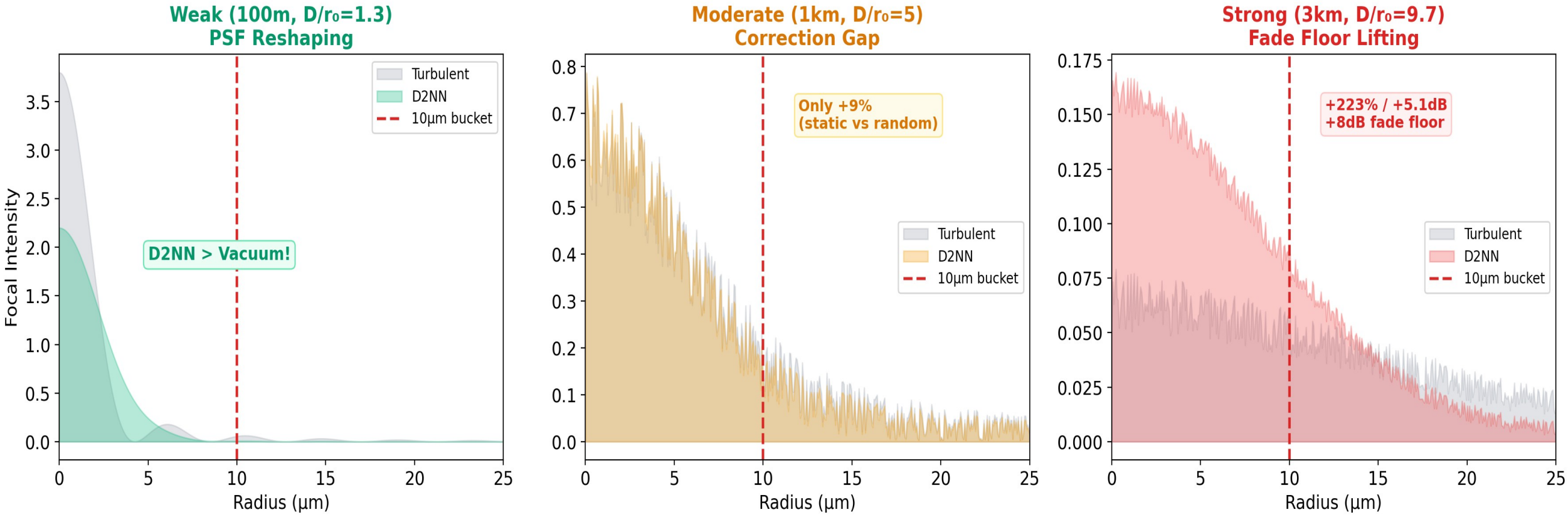
Distance Sweep Results: D2NN Received Power

Static D2NN Performance Across Turbulence Regimes



Distance	D/r_0	σ^{R2}	Vac	Turb	D2NN	Δ (dB)	Fade Δ
100m	1.26	0.015	405.6	382.1	**728.4**	**+2.8**	**+3.2 dB**
500m	3.31	0.28	628.3	499.5	**530.5**	+0.26	+1.3 dB
1km	5.02	1.0	275.1	166.0	**180.9**	+0.37	+2.3 dB

Physical Mechanisms by Turbulence Regime



Static D2NN cannot correct individual wavefronts. It performs ensemble-averaged energy redistribution: PSF reshaping (weak), statistical mode filtering (moderate), deep fade floor lifting (strong).

MPLC vs D2NN: Head-to-Head

	MPLC (Adaptive)	D2NN (Static)
Design	Wavefront matching algorithm	SGD on turbulence ensemble
Operation	Mode decomposition → sorting	Energy redistribution
Adaptivity	**Per-realization (real-time)**	Fixed mask (static)
WF Correction	**Yes**	✘ No (WF RMS unchanged)
Power	Active (sensing + DSP)	**Zero (passive element)**
Complexity	SLM + N detectors + FPGA	**Fixed diffractive element**
Weak Turb	**Strehl → 1.0**	**+91% RP (PSF reshaping)**
Moderate Turb	**Good (mode-limited)**	+6–9% only (correction gap)
Strong Turb	Difficult (scintillation)	**+77–223% (fade mitigation)**
Cost	$$$ (tens of k$)	**$ (one-time fabrication)**

D2NN excels where MPLC struggles (deep turbulence / cost) and vice versa (moderate turbulence / accuracy)

Terminology and Hybrid Architecture

What to Call D2NN?

✘ "Turbulence compensation" (no WF correction)
✘ "Adaptive optics" (static, no feedback)

"Turbulence-resilient passive beam shaping"
"Static diffractive fade mitigation"
"Ensemble-optimized spatial filter"

Proposed Hybrid: D2NN + MPLC

Stage 1 (passive): D2NN reduces fade depth, stabilizes beam
Stage 2 (active): MPLC mode correction on pre-stabilized beam

- D2NN reduces MPLC dynamic range requirement
- Lower MPLC update rate needed
- Graceful degradation if MPLC fails
- Simpler (fewer-mode) MPLC sufficient

+2.8 dB	+5.1 dB	+8.1 dB	0 W	< $100
D2NN Gain (weak)	D2NN Gain (strong)	Fade Floor Improvement	D2NN Power Consumption	D2NN Cost

Key Takeaways

1. D2NN and MPLC share identical physics: cascaded phase masks + free-space propagation = unitary transformation
2. MPLC decomposes into HG modes and corrects adaptively; D2NN applies a fixed, ensemble-optimized spatial filter
3. D2NN is NOT wavefront correction — it's passive turbulence resilience via energy redistribution
4. D2NN advantage: zero power, $-cost, fabricate-once, effective in deep turbulence (+5.1 dB, +8 dB fade floor)
5. MPLC advantage: true per-realization mode correction, full wavefront recovery, but complex and expensive
6. Future: D2NN (passive pre-conditioning) + MPLC (active fine correction) = complementary hybrid architecture